

SHUBHAM MURTADAK

AI-ML Engineer

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ABOUT ME

I am an AI-ML Engineer proficient in Python, Machine Learning, Deep Learning, NLP, Generative AI and Agentic AI passionate about creating impactful solutions. Driven by curiosity and an unwavering desire to learn, I am committed to continuous growth and making meaningful contributions in the field of AI.

EXPERIENCE

Software Engineer — AI&ML

Harbinger Group

Jul 2025 – Present

Pune, Maharashtra, India (Hybrid)

- Worked as an **ML Engineer** on a customer-facing **enterprise HR data governance platform**, developing production-grade **Generative AI** and **LLM-based** solutions for multi-source workforce data onboarding. Built AI modules for automated schema mapping, field discovery, validation rule generation, entity relationship inference, and natural-language policy evaluation, enabling secure, PII-aware knowledge graph ingestion and governance workflows.
- Developed a **Customer Health Pulse analytics platform** using **LangGraph**, **FastAPI**, and **React** to provide leadership stakeholders with insights from **4 years of monthly customer health data**. Implemented **Text-to-SQL retrieval** for natural language querying of structured data and secured the application using **Microsoft Entra ID authentication**. Designed and deployed the full-stack system on an **internal VM**, handling end-to-end architecture, backend APIs, and deployment.
- Developed an **MVP AI-driven QA automation platform** using a **multi-agent architecture** exposed via **MCP servers**. Implemented agents for **test-case generation**, **automation script generation**, and **result reporting** integrated with **Jira** and **Xray Test Management**. Designed an **event-driven pipeline** using **Jira Automation Rules** where workflows are triggered by user-story status changes with **human-in-the-loop validation** before invoking subsequent agents. Built custom **MCP servers** for **Playwright execution** and **Jira/Xray integrations**.
- Developed a **PoC Leave Policy Assistant** using **Azure AI Foundry** to answer employee policy queries via an internal AI agent, exploring agent orchestration and enterprise AI integration.

AI/ML Engineer

Gravitones

Jan 2025 - June 2025

Bhubaneswar, Odisha, India (Remote)

- Develop an AI-powered research paper review automation system for a publication client, integrating Generative AI, NLP, and rule-based logic to automate novelty checks, reference validation, and formatting compliance — reducing manual review effort by 80%, improving accuracy by 40%, and deployed on DigitalOcean.
- Developed an AI-powered cricket fantasy and live match prediction solution for a client, training ML models for real-time win probability on Historical Data, leveraging Generative AI for personalized AI coaching, and Agentic AI for fantasy team and player recommendations — enhancing user engagement, enabling data-driven player trading, and integrating sentiment analysis for accurate match insights.
- Developed a POC AI Healthcare Assistant Agent with Voice-Based Capabilities using LangGraph, capable of interacting with users in real-time audio conversations, handling customer queries using hospital policies, checking and updating records, booking appointments based on available doctors and slots, generating invoices By automating complete end-to-end workflow with 95% accuracy.

Data Science Intern

DataNnoviteSol LLP

June 2024 - jan 2025

Pune, Maharashtra (Onsite)

- Conducted research and development to identify the best tools and solutions for Generative ai project requirements, ensuring optimal fit for various use cases.

- Developed a comprehensive solution for analyzing and processing clients' transaction and campaign data, incorporating Generative AI to optimize campaign performance.
- Develop Generative AI module for *dataRobo*, DataNnovite's AI-driven platform, allowing users to query and retrieve document content via a conversational interface.

SKILLS

Primary Skills:

- **Programming Languages:** Python, R, C++, SQL
- **Python/ML/CV Packages:** Pandas, Numpy, Matplotlib, Seaborn, Scikit-Learn
- **Knowledge of Machine Learning**
- **Deep Learning:** Neural Network, ANN, CNN, Transfer Learning
- **TensorFlow**
- **NLP :** NLTK, Spacy, RNN, LSTM RNN, Transformers
- **Generative AI:** LLM, RAG, Hybrid Rag, **LLM Finetuning**
- LangChain Framework, LlamaIndex Framework, Vector Databases, Graph Databases, VLLM
- **Agentic AI,** Autogen Framework, Crewai Framework ,Langgraph
- **Cloud Platform Services:** AWS :Bedrock, lambda ,ec2 instance, AWS Api Gateway.
- **Microsoft Azure:** AI Foundry (Agent Creation, Finetuning LLMS) ,Azure Openai ,Azure opensearch .
- **Docker**

Secondary Skills:

- **WEB TECHNOLOGIES:** Basics of HTML, CSS, JavaScript, React, Flask, Fastapi, Streamlit
- **DATABASES:** MySQL, MongoDB
- **Visualization Tools:** PowerBI, Microsoft Excel
- **Web Scraping:** Scrapy, BeautifulSoup, Selenium

EDUCATION

B.E. in Artificial Intelligence & Data Science PES Modern College of Engineering Pune CGPA: 9.17	2021-2025
Senior Secondary (XII), Science Shree Ganesh Jr College, HSC Board Grades: 83.38%	2020
Secondary (X) New English School Korhale, India Grades: 89%	2018

PROJECTS

1. DocEase Chat with Documents [Deployed Link](#)

- Developed an AI-powered platform for intelligent interaction with **PDF, PowerPoint, and Word documents**, enabling contextual **question answering** and document analysis using LLMs.
- Implemented **OCR pipelines using AWS Textract and PyTesseract** to process scanned and handwritten documents. Used **AWS Textract** for extracting text from **handwritten documents** and **PyTesseract** for multilingual OCR across **Indian language documents**.

- Built a student learning mode allowing users to upload books or notes (including handwritten PDFs) and generate **summaries, explanations, questions, and MCQ tests** from document content.
- Enabled **multilingual document understanding** across **10 Indian languages** including Marathi, Hindi, Gujarati, Tamil, and Telugu.
- Integrated **text-to-speech and speech-to-text** for voice-based interaction and enabled **document/chat history storage** using Firebase and Firestore.
- **Deployed the full-stack application on AWS EC2** with a custom domain www.docease.in.
- **Tech Stack:** Python | React | FastAPI | LangChain | Gemini-1.5 Pro | AWS Textract | PyTesseract | Milvus | Firebase | Firestore | AWS

2. POExtractor [click here](#)

- Developed an AI tool for real-time email monitoring and automated purchase order (PO) classification.
- Parsed key PO details (PO number, items, quantities, delivery dates, payment terms) using *MiStral-8x7B* and *LLAMA Parser*.
- Built a user-friendly interface with *FastAPI* (backend) and *React* (frontend) for data visualization and manual corrections.
- User can edit and send replies to classified purchase orders mails on React dashboard.
- Enabled multi-format attachment processing (PDF, Excel, CSV, Images, Word).
- **Tech Stack:** Python | LLAMA Parser | LangChain | FastAPI | React

3. Anuwad - English to Marathi Translator [click here](#)

- Developed a neural machine translation model using *TensorFlow*, leveraging the *Bahdanau Attention* mechanism to translate sentences from English to Marathi.
- Implemented an encoder-decoder architecture, where the encoder processes input sequences and the decoder generates target sequences based on context.
- Integrated *Bahdanau Attention* to improve translation quality by enabling the decoder to focus on relevant parts of the input sequence during generation.
- Built a user-friendly UI using *Flask*, allowing users to input English text and receive Marathi translations seamlessly.
- **Tech Stack:** Python | TensorFlow | Bahdanau Attention | Flask

4. RetainIQ [click here](#)

- Developed a system for predicting employee churn using **RandomForest** model to analyze workforce data.
- Implemented features for model training, batch prediction, and single employee prediction to determine retention likelihood.
- Built using Flask for web-based interaction
- **Tech Stack:** Python | SK Learn | ML | Numpy | Pandas | NLP | Flask | GitHub

CERTIFICATES

- LLM Agents MOOC 2024 by universtiy of Berkeley
- Langchain with python bootcamp Udemy
- tensorflow for deep learning Udemy

- Mcp Masterclass: complete Guide to MCP in Python [2025] Udemy
- Masterig Autogen - Building Multiagent Systems Udemy

ACHIEVEMENTS

- Winners Of Prostart 2023(Software Category)-Team Matrix
- Earned Golden Badge in Python On HackerRank